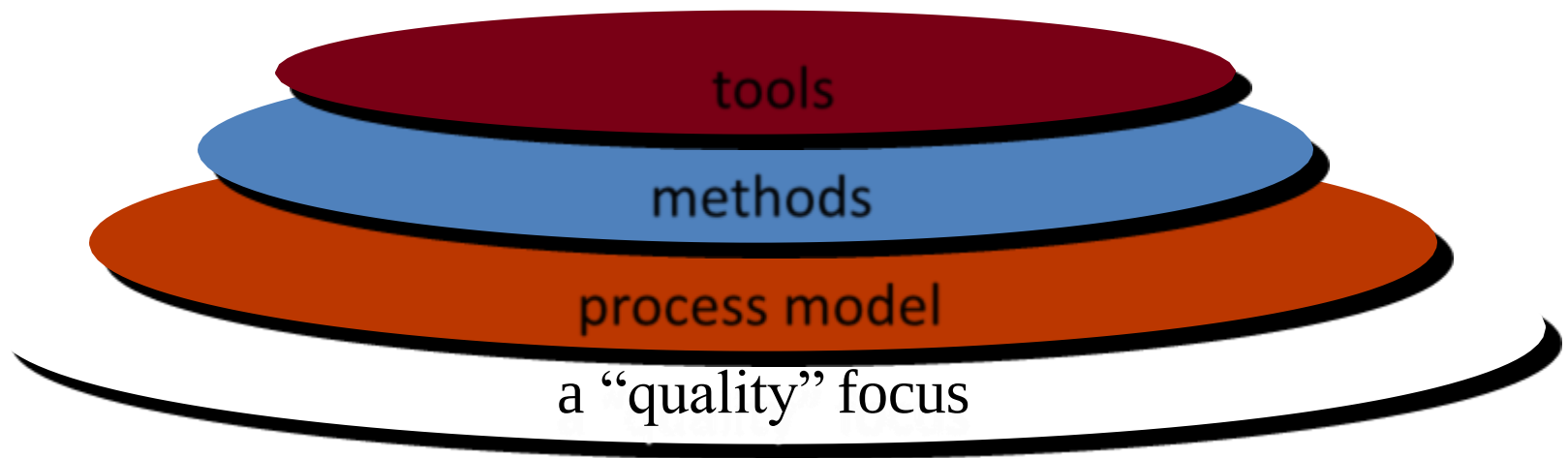


Software Quality Assurance

Software Engineering



Introduction

- **Quality** is defined as a **characteristics or attributes** of something where as attributes refer to measurable characteristics-things that we are able to compare to known standards.
- **There are two kinds of quality:**
 - **Quality of design** refers to the characteristics that designers specify for an item. The grade of materials, tolerances and performance specifications all contribute to the quality of design.
 - **Quality of conformance** is the degree to which the design specifications are followed during manufacturing. Greater the degree of conformance, the higher is the level of quality of conformance

- **Software quality** is defined as conformance to explicitly stated functional and performance requirements, explicitly documented development standards, and implicit characteristics that are expected of all professionally developed software.
- In software development, quality of design encompasses requirements, specifications, and the design of the system where as quality of conformance is an issue focused primarily on implementation.
- If the implementation follows the design and the resulting system meets its requirements and performance goals, conformance quality is high.

Quality Control: Quality control involves the series of inspections, reviews, and tests used throughout the software process to ensure each work product meets the requirements placed upon it. Quality control includes a feedback loop to the process that created the work product. The combination of measurement and feedback allows us to tune the process when the work products created fail to meet their specifications. This approach views quality control as part of the manufacturing process.

Quality control activities may be fully automated, entirely manual, or a combination of automated tools and human interaction

Quality Assurance: Quality assurance consists of the auditing and reporting functions of management. The goal of quality assurance is to provide management with the data necessary to be informed about product quality, thereby gaining insight and confidence that product quality is meeting its goals. Of course, if the data provided through quality assurance identify problems, it is management's responsibility to address the problems and apply the necessary resources to resolve quality issues.

Cost of Quality: The cost of quality includes all costs incurred in the pursuit of quality or in performing quality-related activities. Cost of quality studies are conducted to provide a baseline for the current cost of quality, identify opportunities for reducing the cost of quality, and provide a normalized basis of comparison.

Quality costs may be divided into costs associated with prevention, appraisal, and failure.

Prevention costs include

- quality planning
- formal technical reviews
- test equipment
- training

Appraisal costs include activities to gain insight into product condition the “first time through” each process. Examples of appraisal costs include:

- in-process and inter-process inspection
- equipment calibration and maintenance
- testing

Failure costs are those that would disappear if no defects appeared before shipping a product to customers.

Failure costs may be subdivided into internal failure costs and external failure costs.

Internal failure costs are incurred when we detect a defect in our product prior to shipment. Internal failure costs include

- rework
- repair
- failure mode analysis

External failure costs are associated with defects found after the product has been shipped to the customer. Examples of external failure costs are

- complaint resolution
- product return and replacement
- help line support
- warranty work

SQA Activities

- SQA is composed of a variety of tasks associated with two different constituencies- the software engineer who do technical work and an SQA group that has responsibility for quality assurance planning, oversight , record keeping analysis and reporting.
- The charter of the SQA group is to assist software team in achieving a high quality end product.
- The SEI recommends a set of SQA activities that address quality assurance.

Activities

- **Prepare an SQA plan for a project**
 - The plan is developed during project planning and is reviewed by all interested parties. SQA activities performed by the software engineering team and the SQA team group are governed by the plan. The plan identifies:
 - Evaluations to be performed
 - Audits and reviews to be performed
 - Standards that are applicable to the project
 - Procedures for error reporting and tracking
 - Documents to be produced by the SQA team
 - Amount of feedback provided to the software project team

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- Participates in the development of the project's software process description
 - The software team selects a process for the work to be performed
 - The SQA reviews the process description for compliance with organization policy, internal software standards, externally imposed standards and other parts of software project plan.
- Reviews software engineering activities to verify compliances with defined software process
 - The SQA group identifies, documents and track deviations from the process and verifies that corrections have been made.

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- Audits designated software work products to verify compliance with those defined as part of the software process
 - The SQA reviews selected work products, identifies, documents and track deviations; verifies that correction have been made; and periodically reports the results of its works to the project manager
- Ensures that deviations in software work and work products are documented and handled according to a documented procedures
- Records any noncompliance and reports to senior management

Quality Concepts

- Concerned with ensuring that the required level of quality is achieved in a software product.
- Three principal concerns:
 - At the organizational level, quality management is concerned with establishing a framework of organizational processes and standards that will lead to high-quality software.
 - At the process level, quality management involves the application of specific quality processes and checking that these planned processes have been followed.
 - At the project level, quality management is also concerned with establishing a quality plan for a project. The quality plan should set out the quality goals for the project and define what processes and standards are to be used.

STATISTICAL SOFTWARE QUALITY ASSURANCE

For software, statistical quality assurance implies the following steps:

1. Information about software defects is collected and categorized.
2. An attempt is made to trace each defect to its underlying cause (e.g., nonconformance to specifications, design error, violation of standards, poor communication with the customer).
3. Using the Pareto principle (80 percent of the defects can be traced to 20 percent of all possible causes), isolate the 20 percent (the "vital few").
4. Once the vital few causes have been identified, move to correct the problems that have caused the defects.